

Formative class test - Questions

Visual Perception

Prof Anna Grubert

anna.k.grubert@durham.ac.uk

1. Which of the following is NOT a Gestalt principle?

a) Proximity.

b) Segmentation.

c) Symmetry.

d) Closure.

2. Which statement about ganglion cells is NOT correct?

- a) Magnocellular ganglion cells receive input from rods and are therefore colour-specific.
- b) Magnocellular ganglion cells are also known as parasol cells.
- c) Parvocellular ganglion cells are also known as midget cells.
- d) Parvocellular ganglion cells receive input from L and M cones and are therefore colour-specific.

3. The consequence of the aperture problem is that:
- a) The receptive field of a single neuron never covers the whole visual field.
 - b) The response of a single direction-selective neuron is not sufficient to indicate the motion direction of a stimulus.
 - c) Randomly moving stimuli (motion direction coherence = 0%) do not elicit a response of a single neuron in the middle temporal area.
 - d) Neurons in the middle temporal area respond selectively to a specific motion direction of a stimulus.

4. Mixing lights of different wavelengths to create a new colour hue (e.g., on a computer monitor) is called:

a) Colour-matching.

b) Additive colour mixing.

c) Subtractive colour mixing.

d) Trichromatic colour mixing.

5. What does Richard Gregory (1970) mean with constructed perception?

a) Perception is error-free.

b) Perception is direct.

c) Perception is general.

d) Perception is interpreted.

6. Which of the following keywords is NOT related to the dark adaptation of the eye?

a) Transduction.

b) Rod-cone break.

c) Purkinje effect.

d) Light sensitivity.

7. The psychophysical method of adjustment reveals:

- a) The just noticeable difference between perceived and physical intensity.
- b) The just noticeable conscious experience of a sensory element.
- c) The just noticeable intensity threshold to detect a stimulus.
- d) The just noticeable intensity difference between two stimuli.

8. Which of the following statements about cones, wavelengths and perceived colour is correct?

- a) M and S cones are most sensitive to medium wavelengths of the visible light spectrum. These light waves are perceived as yellow.
- b) S cones are most sensitive to short wavelengths of the visible light spectrum. These light waves are perceived as blue.
- c) M cones are most sensitive to long wavelengths of the visible light spectrum. These light waves are perceived as blue.
- d) M and L cones are most sensitive to short wavelengths of the visible light spectrum. These light waves are perceived as yellow and red.

9. Which of the following statements is NOT correct?
According to the cognitive model of object recognition (Ellis & Young, 1988), objects are recognised when:

- a) Viewer-centred representations are successfully matched against object recognition units.
- b) Object-centred representations are successfully matched against object recognition units.
- c) Perceptual processing stages are matched against object recognition units.
- d) Initial representations are matched against object recognition units.

10. Which of the following is a binocular depth cue?

a) Perspective convergence.

b) Eye Convergence.

c) Motion parallax.

d) Retinal disparity.

11. Which of the following predictions derived from the Corollary Discharge Theory is NOT correct?

- a) If a moving object does not induce retinal motion, the comparator will tell the sensory system that there was no real motion.
- b) If an object induces retinal motion in a moving eye, the comparator will tell the sensory system that there was no real motion.
- c) If an object does not induce retinal motion in a static eye, the comparator will tell the sensory system that there was no real motion.
- d) If an object induces retinal motion in a static eye, the comparator will tell the sensory system that there was real motion.

12. According to the size-distance scaling equation, size constancy depends on:

- a) The size of the retinal image and the physical size of an object.
- b) The physical size and the perceived distance of an object.
- c) The size of the retinal image and the perceived distance of an object.
- d) The perceived size and distance of an object.